LAB 4:

Subnetting and Supernetting using Cisco Packet Tracer

OBJECTIVES:

- To understand the concepts of subnetting and supernetting in computer networks.
- To implement subnetting and supernetting using Cisco Packet Tracer.
- To design basic network structures and assign appropriate IP addresses.
- To verify network connectivity using ping commands.

THEORY:

INTRODUCTION:

Efficient IP address management and proper routing are essential for reliable communication in computer networks. Subnetting and supernetting are important techniques used to optimize IP address utilization and improve routing efficiency. Subnetting divides a large network into smaller segments, while supernetting combines multiple smaller networks into a single larger network.

Subnetting:

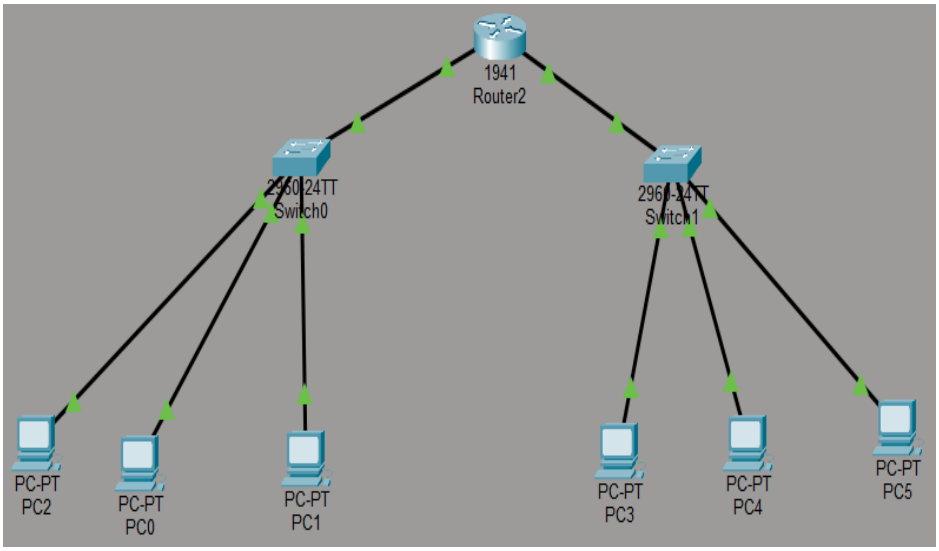
Subnetting is the process of dividing a large IP network into smaller logical subnets by borrowing bits from the host portion of the IP address. This technique improves efficient use of IP addresses, reduces network traffic, and enhances security by separating networks into smaller, manageable units.

Supernetting:

Supernetting is the reverse of subnetting and involves combining two or more contiguous IP networks into one larger network by reducing the number of network bits. This approach simplifies routing, reduces routing table size, and is commonly used in large networks and by internet service providers. Both subnetting and supernetting are essential for effective network design.

NETWORK TOPOLOGY:

1. One Router (1941)
2. Two Switches (2960-24TT)
3. Three pcs per subnet(PC-PT)



NETWORK DESIGN:

Subnetting:

- Base Network Address: 192.168.1.0/24
- Required Number of Subnets: 4
- Borrowed Bits: 2
- New Subnet Mask: /26 (255.255.255.192)
- Number of IP addresses per subnet: 64 (62 usable)

Subnet details:

Subnet 1:

Network ID: 192.168.1.0/26

First Host: 192.168.1.1

Last Host: 192.168.1.62

Broadcast ID: 192.168.1.63

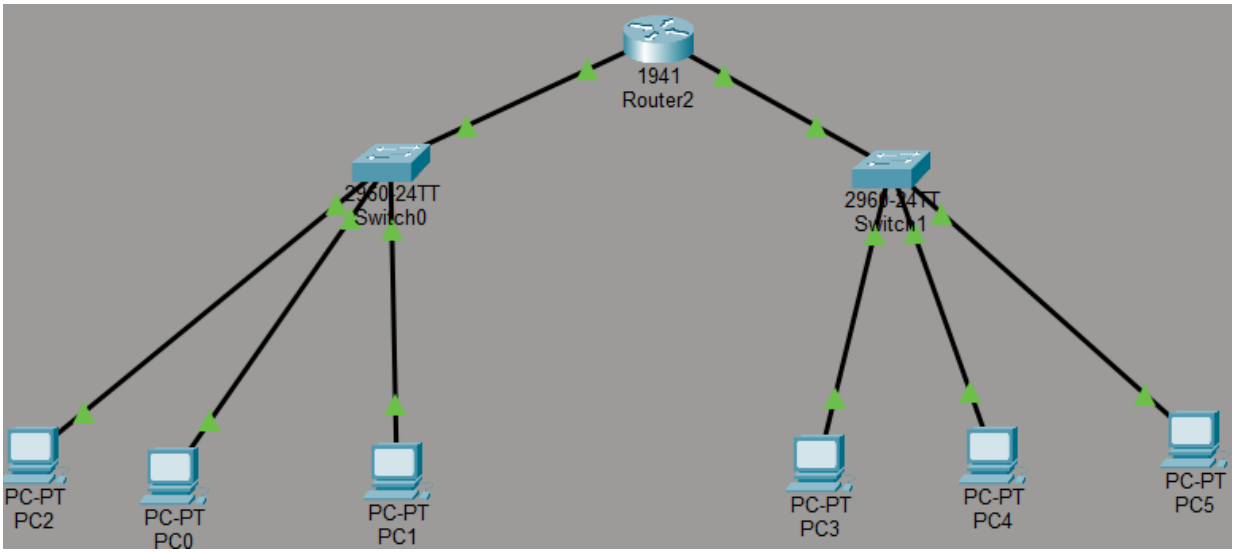
Subnet 2:

Network ID: 192.168.1.64/26

First Host: 192.168.1.65

Last Host: 192.168.1.126

Broadcast ID: 192.168.1.127



Configuration:

For switch one :

Device	IPV4	Subnet Mask	Default gateway
PC-PT S1-PC0	192.168.1.2	255.255.255.192	192.168.1.1
PC-PT S1-PC1	192.168.1.3	255.255.255.192	192.168.1.1
PC-PT S1-PC2	192.168.1.4	255.255.255.192	192.168.1.1

For switch two:

Device	IPV4	Subnet Mask	Default gateway
PC-PT S1-PC0	192.168.1.66	255.255.255.192	192.168.1.65
PC-PT S1-PC1	192.168.1.67	255.255.255.192	192.168.1.65
PC-PT S1-PC2	192.168.1.68	255.255.255.192	192.168.1.65

Network Topology – Supernetting:

Given Networks:

192.168.0.0/24

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

Number of networks: 4 (2^2)

New Supernet Mask: /22 (255.255.252.0)

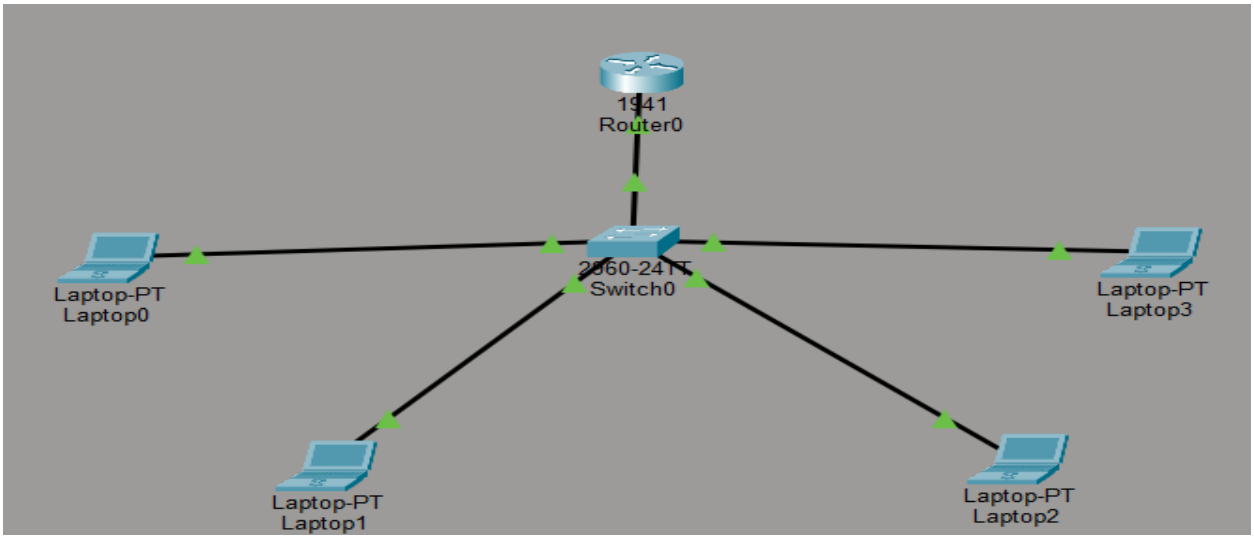
Supernet Address: 192.168.0.0/22

IP Address Range:

First Host: 192.168.0.1

Last Host: 192.168.3.254

Broadcast Address: 192.168.3.255



Configuration:

Device	IPV4	Subnet Mask	Default Gateway
PC-PT S1-PC0	192.168.1.10	255.255.252.0	192.168.1.1
PC-PT S1-PC1	192.168.2.10	255.255.252.0	192.168.1.1
PC-PT S1-PC2	192.168.3.10	255.255.252.0	192.168.1.1
PC-PT S1-PC4	192.168.4.10	255.255.252.0	192.168.1.1

RESULTS:

Subnetting and supernetting were successfully implemented using Cisco Packet Tracer. Proper IP addressing and gateway configuration ensured smooth communication within and across networks. Ping tests confirmed that the configurations were correct.

DISCUSSION AND CONCLUSION:

This experiment successfully demonstrated the practical implementation of subnetting and supernetting using Cisco Packet Tracer. It enhanced understanding of IP addressing, subnet masks, routing concepts, and efficient network design. The successful connectivity tests verified the accuracy of the configurations and reinforced the theoretical concepts.